

Olist E-Commerce Analysis

A student project using Python, SQL and Power BI to understand the Olist Brazilian E-Commerce dataset.

Python → PostgreSQL → Power BI

The project focuses on sales, customers, product categories, delivery performance and reviews.

PROJECT OVERVIEW

For this project, I worked with the Olist Brazilian E-Commerce Public Dataset. The main goal was to understand the marketplace from a few different angles instead of looking at just one metric.

I mainly looked at sales over time, product categories, customer purchasing behaviour, delivery performance and customer reviews. The project was also meant to give me practice with a complete data analyst workflow.

The workflow I followed was:

- Python and pandas for loading, cleaning and checking the data.
- PostgreSQL and SQL for the main analysis.
- Power BI for the dashboard and visual presentation.

DATASET

The dataset contains information about orders, customers, products, sellers, payments, reviews, category translations and geolocation. The orders table contains 99,441 orders, while the order-items table contains 112,650 item records.

One important thing I had to keep in mind was that an order can contain multiple items. Because of this, revenue analysis was based on the price from the order-items table.

DATA CLEANING

Before doing the analysis, I checked the tables for missing values, duplicate rows, data types and identifier problems. I also converted the order and delivery date columns into datetime values.

Some of the main decisions I made were:

- Missing product categories were kept as unknown instead of deleting those products.
- Missing review comments were kept because a review does not have to contain a written comment.
- Review IDs were checked rather than automatically assuming every ID represented one unique order.
- Delivery-date anomalies were investigated and only excluded when they affected delivery calculations.

SALES PERFORMANCE

The analysis of delivered orders gave total delivered sales of about R\$13.22 million across 96,478 delivered orders. The average order value was R\$137.04.

Sales generally increased during 2017. There were some month-to-month decreases as well, so the trend was not a straight line. November 2017 had the highest monthly delivered sales in the dataset.

I noticed that the earliest months had very small numbers of orders. Because of that, some percentage growth values for those months were extremely large. I did not treat those early percentage changes as meaningful business trends.

PRODUCT CATEGORIES

Looking at delivered revenue by category, health & beauty was the highest-revenue category. It was followed by watches & gifts and bed & bath table.

Other categories that appeared strongly in the analysis included sports & leisure, computers & accessories and furniture & decor.

CUSTOMER BEHAVIOR

Around 97% of customers made only one delivered purchase during the available observation period, while around 3% made more than one.

The repeat-customer group was much smaller, but repeat customers had higher average spending per customer. This should be interpreted with the limited observation period in mind.

DELIVERY PERFORMANCE

Average delivery time was approximately 12.5 days. Among the valid delivered orders used for this analysis, 8.11% were delivered later than the estimated delivery date.

The late-delivery rate was not the same across all states. I used a minimum order threshold so states with very few orders would not distort the comparison.

DELIVERY AND CUSTOMER REVIEWS

I compared review scores between late and on-time deliveries. Late deliveries had an average review score of 2.57, while on-time deliveries had an average score of 4.29.

This shows a strong association between delivery performance and review scores in the dataset, but it does not prove that late delivery itself caused the lower score.

MAIN FINDINGS

- Delivered sales were about R\$13.22M across 96,478 delivered orders.
- Average order value was R\$137.04.
- Health & beauty generated the highest delivered revenue.
- About 97% of customers made only one delivered purchase.
- About 8.11% of valid delivered orders were late.
- Late deliveries had a much lower average review score than on-time deliveries.

LIMITATIONS

This is a historical dataset, so the results should not be treated as a description of Olist's current performance. The repeat-customer percentage is also affected by the available observation period.

The delivery and review comparison shows a relationship in the data, but it does not prove that late delivery was the only reason for lower review scores.